

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

THE MULTINOMIAL DISTRIBUTION

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

The Multinomial Distribution

Imagine a k —faced die, with probabilities $p_1, \dots, p_k$.

Toss such a die n times, and count the number of times each of the k faces occurs:

$$X_j = \# \text{ of times face } j \text{ occurs}$$

The distribution of $X = (X_1, \dots, X_k)$ is called the **multinomial**.

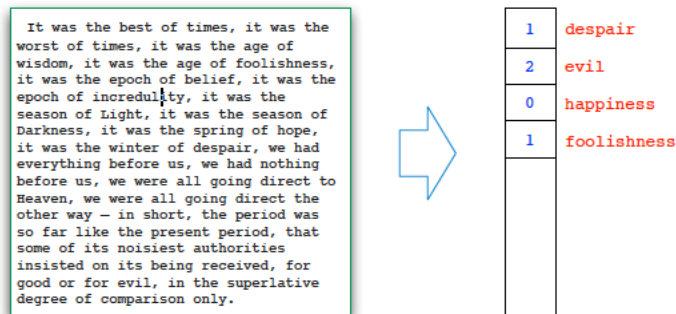
- Parameters: $p_1, \dots, p_k \geq 0$, with $p_1 + \dots + p_k = 1$.
- $\mathbb{E}X = (np_1, np_2, \dots, np_k)$.
- $\Pr(n_1, \dots, n_k) = \binom{n}{n_1, n_2, \dots, n_k} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}$, where

$$\binom{n}{n_1, n_2, \dots, n_k} = \left\{ \frac{n!}{n_1! n_2! \dots n_k!} \right\},$$

the # of ways to place balls numbered $\{1, \dots, n\}$ into bins numbered $\{1, \dots, k\}$.

Example: Text Documents

Bag-of-words: vectorial representation of text documents.



- Fix V = some vocabulary.
- Treat words in document as independent draws from a multinomial over V :

$$p = (p_1, \dots, p_{|V|}), \text{ such that } p_i \geq 0 \text{ and } \sum_i p_i = 1$$

How would we estimate the parameters of a multinomial?